

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy at Project Check-in.

Your abstract should be written after you finish the research and experimentation and should include:

- **Project number**, title and the full name of every team member
- Purpose of the project
- Procedure
- Observations/data/results
- Conclusions
- 250 words or less (to fit on one page)
- Use a black font size 10 to 12

I48

A Low-Cost, AI-Powered Safety Device to Prevent Child Hot Car Deaths

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Purpose: According to Kids and Car Safety, about 34 kids die each year after being left locked in a hot car. The purpose is to design a low-cost safety device that can prevent hot car deaths.

Materials/Procedure: My low-cost safety device is built using a Raspberry Pi Zero-W computer with a camera accessory to capture live video, DHT22 temperature sensor, a USB speaker and a battery power bank. The device is enclosed in a compact 3D-printed plastic enclosure with holes for ports and camera, and a mounting kit for easy car mounting.

The brains of the device is my Python software using an AI/ML model to continuously detect one or more children left alone in the car and if the detected temperature rises beyond the threshold, plays a loud audio alert message through its built-in USB speaker prompting passersby to respond. The entire device costs less than \$100.

Observations: I mounted my device in our car on the inside windshield facing the backseat. I set its threshold to 1 child and 75°F temperature. I tested various combinations of children in the car and temperature settings. In all cases, the device functioned as expected, sounding alerts only when there was a single child and inside temperature exceeded 75°F.

Conclusion: We demonstrated a cost-effective, fully self-contained device mountable in any car, that can accurately detect and alert a lone child in a hot car preventing harm.